Smartly — Service and Data Design

Project: Access Flow Control Management System for classrooms, labs and shared resources.

Service ownership: Identity owns users/roles; Resource Catalogue owns buildings/rooms/equipment; Booking owns requests and time conflict rules; Approval owns decisions/audit; Notification owns preferences and delivery attempts.

Contract rule: services communicate using documented operations; no service reads another service's tables directly. The architecture is a logical design, while the current implementation is a single Flask prototype with in-memory storage.

Central flow: a student searches rooms, submits a booking, the Booking service validates and checks overlap, faculty approve/reject, and Notification sends a decision notice.

Schema: Student, Building, Room, Resource, Booking and Decision. Primary/foreign key fields are shown in schema.drawio/schema.svg; database/schema.sql is the implementation-oriented DDL.

Quality checks: reachable endpoints, self-contained ownership, published contracts, independent versioning/deployment, and loose coupling.